

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

ME 19-205-0405 HYDRAULIC MACHINERY (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand dimensional analysis and principles of similitude.
 CO2: Apply the momentum principles of a jet and analyse its performance.
 CO3: Learn the working of various hydraulic turbines and their performance characteristics.
 CO4: Understand the working of roto dynamic pumps and positive displacement pumps.
 CO5: Study the principle of working of hydraulic devices.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
1. (a) Prove that the force exerted by a jet of water on a fixed semi-circular plate in the direction of the jet when the jet strikes at the centre of the semi-circular plate is two times the force exerted by the jet on an fixed vertical plate.	3	L3	1	1,2	
(b) In 1 in 60 model of a spillway, the velocity and discharge are 3 m/s and 3 m ³ /s. Find the corresponding velocity and discharge in the prototype.	3	L2	2	1,2	
(c) What is priming? Why is it necessary?	3	L1	4	1	
(d) Explain the principle and working of a Kaplan turbine?	3	L1	3	1,2	
(e) Define specific speed of a turbine. Derive an expression for the same?	3	L1	3	1,2	
(f) A single-acting reciprocating pump running at 30 rpm delivers 0.012 m ³ /s of water. The diameter of the piston is 25 cm and stroke length is 50 cm. Determine: (i) The theoretical discharge of the pump. (ii) Coefficient of discharge. (iii) Slip and percentage slip of the pump.	3	L2	4	1,2	
(g) Explain the principle and working of a hydraulic press.	3	L1	5	1	
(h) Explain the principle and working of hydraulic intensifier.	3	L1	5	1,2	

PART B

(4 × 12 = 48)

- II. (a) Determine an expression for force exerted by a jet of water on a fixed vertical plate in the direction of jet. 4 L3 2 2,3
- (b) A jet of water moving at 16 m/s strikes a curved vane which is moving with a velocity of 5 m/s. The vane is symmetrical and is so shaped that the jet is deflected through an angle 120°. Find the angle of jet at inlet of the vane so that there is no shock at inlet. What is the absolute velocity of jet at outlet in magnitude and direction and work done per unit weight of water striking per second? Assume that the vane is smooth. 8 L3 2 3,4

OR

(P.T.O.)

		Marks	BL	CO	PO
III.	(a) Water is flowing through a pipe of diameter 40 cm at a velocity of 3 m/s. Find the velocity of oil flowing in another pipe of diameter 10 cm, if the condition of dynamic similarity is satisfied between the two pipes. The viscosity of water and oil is given as 0.011 Poise and 0.026 Poise. The specific gravity of oil is 0.8.	6	L3	1	1,2,3
	(b) The variables controlling the motion of a floating vessel through water are the drag force F , the speed V , the length L , the density ρ , the dynamic viscosity μ of water and acceleration due to gravity g . Derive an expression for F by dimensional analysis.	6	L3	1	1,2,3
IV.	(a) Define the following efficiencies of a turbine: (i) Hydraulic efficiency. (ii) Mechanical efficiency. (iii) Volumetric efficiency. (iv) Overall efficiency.	4	L3	3	3,4
	(b) A Pelton wheel has a mean bucket speed of 10 m/s with a jet of water flowing at the rate of $0.8 \text{ m}^3/\text{s}$ with a head of 32 meters. The buckets deflect the jet through an angle of 160° . Calculate the power given by water to the runner and the hydraulic efficiency of the turbine. Assume coefficient of velocity is 0.985.	8	L3	3	3,4
OR					
V.	(a) What is a draft tube and why is it used in a reaction turbine? Describe with neat sketch two different types of draft tubes.	6	L3	3	3,4
	(b) A conical draft-tube having diameter at the top as 2.0 m and pressure head at 7 m of water (vacuum), discharges water at the outlet with a velocity of 1.3 m/s at the rate of $25 \text{ m}^3/\text{s}$. If atmospheric pressure head is 10.3 m of water and losses between the inlet and outlet of the draft tubes are negligible, find the length of draft-tube immersed in water. Total length of tube is 5 m.	6	L3	3	3,4
VI.	(a) Explain the working of a centrifugal pump with a neat sketch and define the following terms: (i) suction head (ii) delivery head (iii) static head (iv) manometric head.	4	L3	4	3,4
	(b) A centrifugal pump having outer diameter equal to two times the inner diameter and running at 1000 rpm works against a total head of 40 m. The velocity of flow through the impeller is constant and equal to 2.5 m/s. The vanes are set back at an angle of 40° at outlet. If the outer diameter of impeller is 500 mm and width at outlet is 50 mm, determine: (i) Vane angle at inlet. (ii) Work done by impeller on water per second. (iii) Manometric efficiency.	8	L3	4	3,4

OR

(Continued)

		Marks	BL	CO	PO
VII.	(a) Define slip, percentage slip and negative slip of a reciprocating pump.	4	L3	4	3,4
	(b) The diameter and stroke length of a single-acting reciprocating pump are 100 mm and 200 mm respectively. The lengths of suction and delivery pipes are 10 m and 30 m respectively and their diameters are 50 mm. If the pump is running at 30 rpm and suction and delivery heads are 3.5 m and 20 m respectively, find the pressure head in the cylinder:	8	L3	4	3,4
	(i) At the beginning of the suction and delivery stroke.				
	(ii) In the middle of suction and delivery stroke.				
	(iii) At the end of suction and delivery stroke.				
	Take atmospheric pressure head = 10.30 meters of water and coefficient of friction = 0.0089 for both pipes.				
VIII.	Explain the principle and working of the following hydraulic devices with the help of a neat sketch	12	L2	5	4,5
	(i) Hydraulic press.				
	(ii) Hydraulic ram.				
OR					
IX.	Explain the principle and working of the following hydraulic devices with the help of a neat sketch	12	L2	5	4,5
	(i) The gear-wheel pump.				
	(ii) Hydraulic crane.				

Blooms's Taxonomy Level's

L1 - 12.5%, L2 - 25%, L3 - 62.5%.
